PROJECT PLAN

NEXUS ERP - Integrated ERP Portal with Employee Risk Scoring

Full Project Timeline & Execution Plan

Project:	NEXUS ERP (Package D - Full Package)
Client:	Nethmina Kavinda Bandara
Total Budget:	LKR 80,000
Start Date:	31st March 2026
Delivery Date:	13th May 2026
Duration:	~7 weeks (46 days)

Internal planning document - Not for client distribution

Context & Constraints

- Available hours: 4-5 hours/day (1-2 other client projects running in parallel)
- Total estimated work hours: ~190 hours across 46 days
- SLR submission deadline: 10th April 2026 (URGENT)
- Client second payment (LKR 20,000): 14th April 2026
- Mentorship sessions: 4 free sessions post-delivery
- Research writing and development must run in sequence (SLR first, then dev)

Summary Timeline

Week	Phase	Priority	Hours
Week 1 (Apr 1-10)	SLR Writing + Submission	CRITICAL	43.5
Week 2 (Apr 11-15)	Backend Core (Auth, Employee, Attendance)	HIGH	22.5
Week 3 (Apr 16-20)	Backend (Payroll, Inventory, Social Score)	HIGH	21
Week 4 (Apr 21-25)	ML Microservice (RF, SHAP, Feedback)	CRITICAL	22.5
Week 5 (Apr 26-May 2)	Frontend (All pages + Dashboard)	MEDIUM	31.5
Week 6 (May 3-10)	Evaluation + Thesis Writing	CRITICAL	36.5
Week 7 (May 11-16)	Testing + Delivery + Mentorship	HIGH	12

Hour Budget Breakdown

Phase	Days	Hours	% of Total
Phase 1: SLR	10	43.5	23%
Phase 2: Backend	10	43.5	23%
Phase 3: ML Microservice	5	22.5	12%
Phase 4: Frontend	7	31.5	17%
Phase 5: Evaluation	3	13	7%
Phase 6: Thesis Writing	5	23.5	12%
Phase 7: Testing/Polish	2	9	5%
Phase 8: Delivery + Mentorship	4	3	1%
TOTAL	46	~190	100%

Phase 1: SLR + Research Foundation (Apr 1-10)

CRITICAL SLR submission April 10

Week 1 (Apr 1-6): SLR Writing

Day	Task	Hrs	Output
Apr 1	Paper search - 20-25 papers across 5 SLR themes	4.5	Paper list
Apr 2	Read & summarise 8-10 core papers (ML, RF, social)	4.5	Summary notes
Apr 3	Read remaining papers (ERP, synthetic, XAI, security)	4.5	All summarised
Apr 4	Draft SLR - Intro, search strategy, criteria	4.5	Sections 1-2
Apr 5	Draft themes 1-2: ERP systems, ML attrition	4.5	Themes 1-2
Apr 6	Draft themes 3-5: Social signals, synthetic data, XAI	4.5	Themes 3-5

Week 1 cont. (Apr 7-10): SLR Completion

Day	Task	Hrs	Output
Apr 7	Comparison table, gap analysis section	4.5	Gap identified
Apr 8	Research gap justification, link to novelty	4.5	Gap complete
Apr 9	Full review, IEEE formatting, references	4.5	SLR draft done
Apr 10	Final proofread, deliver to client	3	SLR SUBMITTED

Deliverable	**SLR document ready for submission**

Phase 2: Backend Development (Apr 11-20)

HIGH Build data infrastructure for ML model

Strategy: Build backend first because (1) ML model needs attendance/payroll/social data to train on, (2) seeding scripts populate the database, (3) frontend can be built against working APIs later.

Week 2 (Apr 11-15): Core Backend

Day	Task	Hrs	Output
Apr 11	Project scaffold - Node.js, Express, MongoDB, .env	4.5	Boilerplate running
Apr 12	Mongoose schemas - all 9 collections	4.5	Schemas defined
Apr 13	Auth - JWT, bcrypt, role middleware, OTP	4.5	Auth working
Apr 14	Employee CRUD + seeding (20 employees, risk_bias)	4.5	Employee API + seed
Apr 15	Attendance module + seeding (6 months data)	4.5	Attendance API + seed

Week 3 (Apr 16-20): Payroll, Inventory, Social Score

Day	Task	Hrs	Output
Apr 16	Payroll module - auto-calc from attendance	4.5	Payroll API
Apr 17	Payroll seeding + inventory module start	4.5	Payroll seed + inv start
Apr 18	Inventory completion - alerts, seed data	4.5	Inventory complete
Apr 19	Social Score seeding - Gaussian, Box-Muller, drift	4.5	socialScoreSeed.js
Apr 20	Feature vector construction - 8 features per emp	4.5	Features populated

Deliverable	**Full backend API with all seed data**

Phase 3: ML Microservice (Apr 21-25)

CRITICAL Core research contribution

Day	Task	Hrs	Output
Apr 21	Flask scaffold - /predict, /train, /health, CORS	4.5	Flask running
Apr 22	Random Forest - 200 trees, train, predict, joblib	4.5	RF model serving
Apr 23	Weighted composite scoring in Node.js, WRL calc	4.5	Dual scoring path
Apr 24	HR feedback loop - accept/reject, retrain endpoint	4.5	Feedback loop done
Apr 25	SHAP integration - explainer, feature importance	4.5	SHAP working

Deliverable **Full ML pipeline - predict, train, retrain, explain**

Phase 4: Frontend Development (Apr 26-May 2)

MEDIUM UI for demonstration and thesis screenshots

Day	Task	Hrs	Output
Apr 26	React scaffold - Vite, routing, auth pages, JWT	4.5	App shell + auth
Apr 27	Employee pages - list, profile, CRUD forms	4.5	Employee UI
Apr 28	Attendance pages - check-in/out, monthly summary	4.5	Attendance UI
Apr 29	Payroll pages - list, payslip view, generation	4.5	Payroll UI
Apr 30	Inventory pages - stock list, transactions, alerts	4.5	Inventory UI
May 1	Dashboard - KPIs, risk charts, WRL, dept breakdown	4.5	Main dashboard
May 2	Risk scoring UI - SHAP viz, accept/reject, retrain	4.5	Risk UI complete

Deliverable **Full frontend - all pages + dashboard + risk UI**

Phase 5: Evaluation + ML Analysis (May 3-5)

CRITICAL Produces thesis Chapter 5 data

Day	Task	Hrs	Output
May 3	ML eval - confusion matrix, precision/recall/F1, CV	4.5	ML metrics
May 4	SHAP plots + social score validation (histograms, Q-Q)	4.5	Charts generated
May 5	Retraining comparison + functional testing	4	Full eval data

Deliverable	All evaluation metrics, charts, plots

Phase 6: Thesis Writing (May 5-10)

HIGH Complete thesis document

SLR (Chapter 2) already done. Focus on Chapter 4 (Design - ML deep dive) and Chapter 5 (Evaluation results).

Day	Task	Hrs	Output
May 5	Chapter 1 - Introduction (refine from proposal)	1	Ch.1 draft
May 6	Chapter 3 - Methodology (DSR, DSRM, ethics)	4.5	Ch.3 complete
May 7	Chapter 4 - ERP architecture, DB, API design	4.5	Ch.4 part 1
May 8	Chapter 4 - Social score, risk pipeline, ML, feedback	4.5	Ch.4 part 2
May 9	Chapter 5 - ML results, SHAP, validation, testing	4.5	Ch.5 complete
May 10	Chapter 6 - Conclusion, formatting, IEEE refs, TOC	4.5	THESIS COMPLETE

Deliverable	Complete thesis (Chapters 1-6)

Phase 7: Integration Testing + Polish (May 11-12)

Day	Task	Hrs	Output
May 11	E2E testing - full workflow, bug fixes	4.5	System tested
May 12	Code docs, README, GitHub repo, thesis proofread	4.5	Everything polished

Phase 8: Delivery + Mentorship (May 13-16)

Day	Task	Hrs	Output
May 13	FINAL DELIVERY - source code + thesis + docs	1	Project delivered
May 13	Session 1 - Project overview, ERP walkthrough	0.5	Session done
May 14	Session 2 - Risk scoring, features, ML model	0.5	Session done
May 15	Session 3 - Backend deep dive, APIs, auth, DB	0.5	Session done
May 16	Session 4 - Frontend walkthrough, React, state	0.5	Session done

Payment Milestones

Date	Amount (LKR)	Trigger	Status
31 Mar 2026	10,000	Advance payment	PAID
14 Apr 2026	20,000	Client salary day	Pending
13 May 2026	50,000	Final delivery	Pending
Total Project Value		**LKR 80,000**	

Risk Mitigation

Risk	Mitigation Strategy
SLR takes longer than expected	Focus on abstracts + methodology + results, not full papers
Backend falls behind	Use boilerplate CRUD patterns. Seed scripts > perfect APIs
ML model underperforms	Synthetic data has known ground truth. Tune hyperparameters if needed
Frontend takes too long	Use component library (MUI/Ant). Dashboard is priority page
Thesis writing crunch	Ch.2 done early. Ch.3 is short. Buffer time for Ch.4-5
Client requests scope changes	Agreement states scope changes = revised pricing

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This is an internal planning document. Execute according to the daily schedule.

Refer to SOURCE_OF_TRUTH.md for full technical specifications.